

BANKING DASHBOARD PROJECT

Solution Document — Power BI Dashboards & Risk Analytics

1. Solution Overview

Interactive Power BI dashboards were developed to help the bank make lending decisions based on the applicant's profile. If the applicant is likely to repay the loan, the loan can be approved; otherwise it can be declined. The solution presents key metrics for loans, deposits, client engagement, and fees, with rich filtering by Joining Year, Gender, Banking Relationship, and Institution Advisor.

2. Key Calculated Functions (DAX)

SUM — Adds all numbers in a column.

```
Bank Deposit = SUM('Clients - Banking'[Bank Deposits])
```

DISTINCTCOUNT — Counts distinct values in a column.

```
Total Clients = DISTINCTCOUNT('Clients - Banking'[Client ID])
```

SUMX — Sum of an expression evaluated for each row.

```
Total Fees = SUMX('Clients - Banking', [Total Loan] * 'Clients - Banking'[Processing Fees])
```

DATEDIFF — Number of interval boundaries between two dates.

```
Engagement Days = DATEDIFF('Clients - Banking'[Joined Bank], TODAY(), DAY)
```

SWITCH — Evaluates an expression against a list of values and returns corresponding results.

3. Key Performance Indicators (KPIs)

KPI	Description / Formula
Total Clients	DISTINCTCOUNT of Client ID — total number of clients.
Total Loan	Bank Loan + Business Lending + Credit Cards Balance.
Bank Loan	SUM of Bank Loans — amount to be repaid by clients.
Business Lending	SUM of Business Lending — loans to small businesses.
Total Deposit	Bank Deposit + Savings + Foreign Currency + Checking Accounts.
Total Fees	SUMX of Total Loan × Processing Fees.
Bank Deposit	SUM of Bank Deposits.
Checking Accounts	SUM of Checking Accounts — daily transactional access.
Savings Account	SUM of Saving Accounts — interest-bearing deposits.
Foreign Currency Account	SUM of Foreign Currency Account balances.
Credit Cards Balance	SUM of Credit Card Balance currently owed.
Engagement Length	SUM of Engagement Days — client tenure with the bank.

4. Visualization — Home Dashboard

The Home page provides a high-level snapshot of the banking portfolio with key cards for Total Clients, Total Loan, Total Deposit, Checking Accounts, Savings Accounts, and Business Lending. Filters for Joining Year and Gender allow quick slicing of the data. Navigation buttons lead to Loan Analysis, Deposit Analysis, and Summary views.

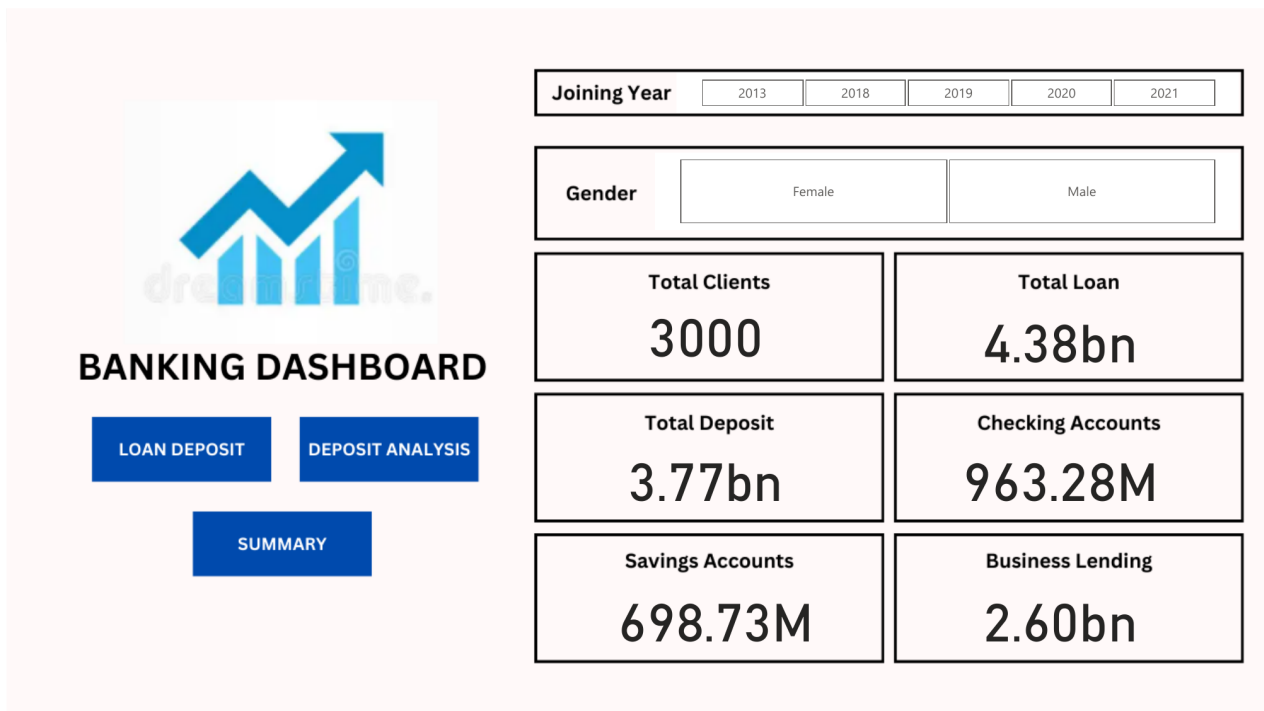


Figure 1: Banking Dashboard — Home View

Key insights from Home: 3,000 clients with Total Loan of 4.38 bn and Total Deposit of 3.77 bn. Business Lending forms a large share (2.60 bn) of the loan book. Checking and Savings accounts contribute significantly to deposits.

5. Visualization — Loan Analysis

The Loan Analysis page breaks down Total Loan, Bank Loan, Business Lending, and Credit Card Balance. Visuals include Bank Loan by Banking Relationship (BR), by Occupation, by Income Band (donut), and by Nationality. Slicers for Banking Relationship, Gender, and Institution Advisor support detailed risk and exposure analysis.

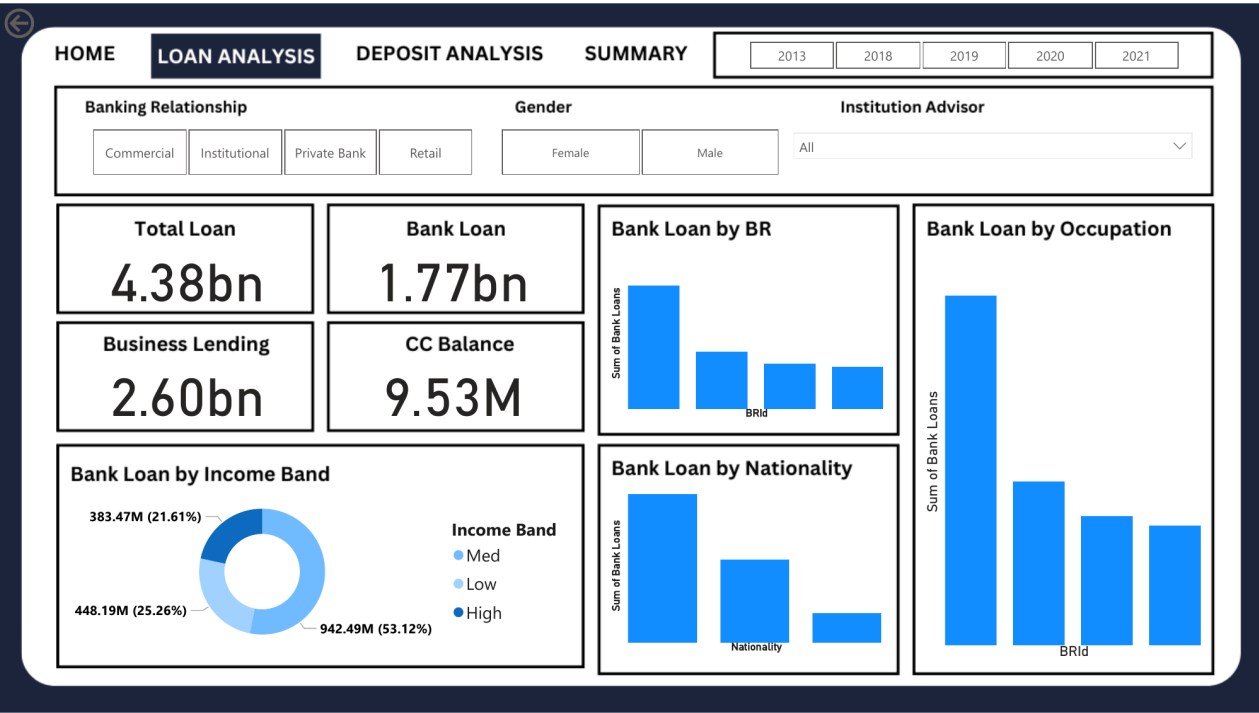


Figure 2: Loan Analysis Dashboard

Key insights from Loan Analysis: Total Loan 4.38 bn (Bank Loan 1.77 bn + Business Lending 2.60 bn + CC Balance 9.53 M). High Income Band accounts for the largest share of Bank Loans (~53%). Distribution by Banking Relationship and Occupation highlights concentration risk that can be monitored for lending decisions.

6. Visualization — Deposit Analysis

The Deposit Analysis page focuses on Total Deposit, Bank Deposit, Saving Account Amount, and Checking Account Amount. Charts show Bank Deposit by Banking Relationship, by Occupation, by Income Band, and by Nationality, enabling understanding of the deposit base and client segments that contribute most to funding.

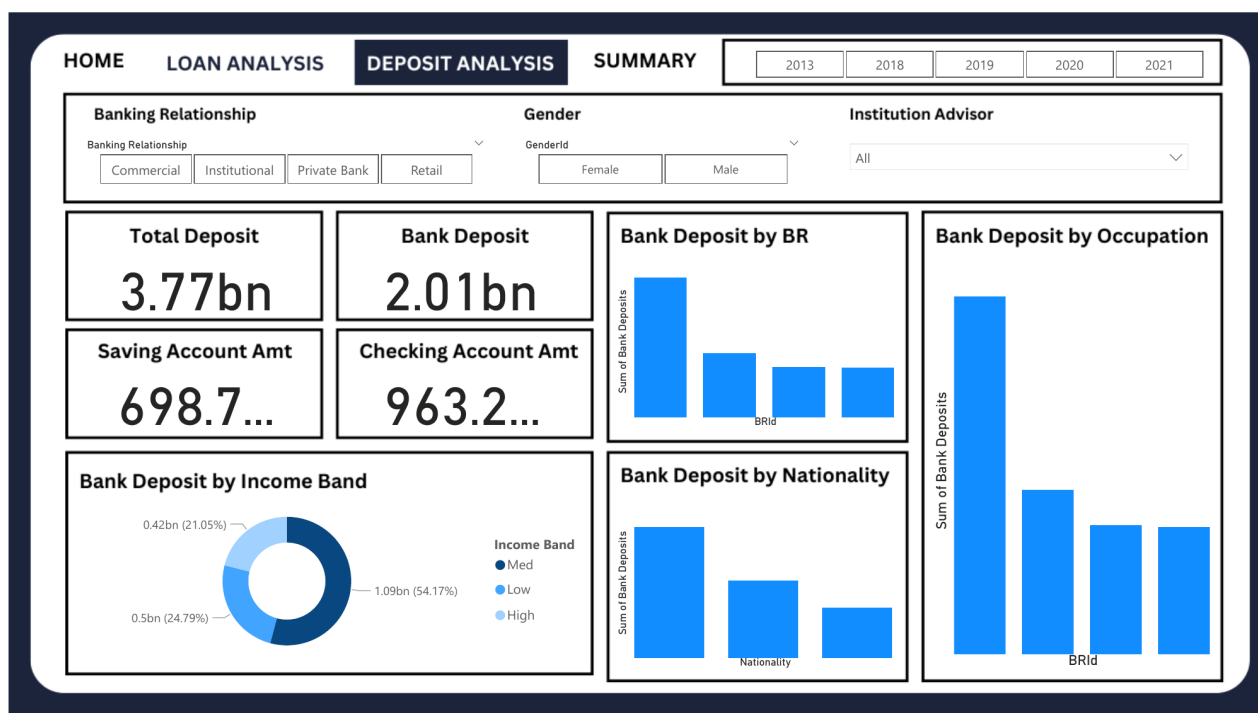


Figure 3: Deposit Analysis Dashboard

Key insights from Deposit Analysis: Total Deposit 3.77 bn (Bank Deposit 2.01 bn). High Income Band dominates deposits (~54%). Patterns by BR, Occupation, and Nationality help the bank design products and retention strategies for high-value depositors.

7. Conclusion

Empowered by modern data visualization techniques, Power BI dashboards are highly effective in the banking sector. A banking operations dashboard built with key metrics and KPIs enables better risk assessment, faster decisions, and clearer visibility into loan and deposit portfolios.

8. Future Work & Recommendations

- Use the dashboards to monitor total loan amount and related metrics by investor / advisor.
- Analyse which bank relationship types (e.g., Private Bank) attract more clients and refine acquisition strategies.
- Identify nationalities and income bands with highest bank loans for targeted risk review.
- Extend analysis to more granular product-level and time-series trends for proactive risk management.